Quiz 3

Name: _____

AI 501: Mathematics for AI | Duration: 20 min | Student ID: _____
Total: 10 marks

Question 1 (2 marks)

Decompose the following matrix A as $A = S + K$, where S is symmetric and K is skew-symmetric.

$$A = \begin{pmatrix} 4 & 2 \\ 0 & 2 \end{pmatrix}.$$

Solution:

$$A^T = \begin{pmatrix} 4 & 0 \\ 2 & 2 \end{pmatrix}, \qquad S = \frac{A + A^T}{2} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \qquad K = \frac{A - A^T}{2} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Question 2 (2 marks)

Given a matrix A and a vector $\mathbf{b}$ as follows:

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 4 & -2 \\ 0 & 5 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 7 \\ 3 \\ 8 \end{pmatrix}$$

Compute the product $A\mathbf{b}$ as a linear combination of the columns of A.

Solution: To compute the product $A\mathbf{b}$, as a linear combination of the columns of A:

$$A\mathbf{b} = 7 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} + 8 \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$$

Question 3 (6 marks)

Consider two vectors in $\mathbb{R}^3$

$$\mathbf{u}_2 = (1, 0, 1), \quad \mathbf{u}_1 = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right).$$

(a) [1 mark] Describe $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ geometrically in $\mathbb{R}^3$.

(b) [5 marks] Apply the Gram–Schmidt process to the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ to produce an orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$.
Hint: If one of $\mathbf{u}_1$ or $\mathbf{u}_2$ already has unit norm, choose it as $\mathbf{v}_1$.

Solution:

1. The span of these two vectors is

$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2\} = \{\alpha\mathbf{u}_1 + \beta\mathbf{u}_2 : \alpha, \beta \in \mathbb{R}\}.$$

Since $\mathbf{u}_1$ and $\mathbf{u}_2$ are not scalar multiples of each other, they are linearly independent. **Therefore, their span is a 2-dimensional subspace of $\mathbb{R}^3$, i.e., a plane through the origin.**

Moreover, the plane spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$ has normal vector

$$\mathbf{n} = \mathbf{u}_1 \times \mathbf{u}_2 = \left(\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}\right).$$

Hence, the equation of the plane is

$$x - y - z = 0.$$

$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2\} = \{(x, y, z) \in \mathbb{R}^3 : x - y - z = 0\}.$$

2.

$$\|\mathbf{u}_1\|^2 = \left(\tfrac{1}{\sqrt{2}}\right)^2 + \left(\tfrac{1}{\sqrt{2}}\right)^2 + 0^2 = \tfrac{1}{2} + \tfrac{1}{2} = 1.$$

So $\mathbf{u}_1$ is unit-normalized. Now Orthogonalize $\mathbf{u}_2$ with respect to $\mathbf{u}_1$.

$$\mathbf{v}_2 = \mathbf{u}_2 - \langle \mathbf{u}_2, \mathbf{u}_1 \rangle \mathbf{u}_1.$$

$$\langle \mathbf{u}_2, \mathbf{u}_1 \rangle = (1,0,1) \cdot \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) = \tfrac{1}{\sqrt{2}}.$$

So,

$$\mathbf{v}_2 = (1,0,1) - \tfrac{1}{\sqrt{2}} \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) = (1,0,1) - \left(\tfrac{1}{2}, \tfrac{1}{2}, 0\right) = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right).$$

Now normalize the result to obtain an orthonormal basis.

$$\|\mathbf{v}_2\| = \sqrt{\left(\tfrac{1}{2}\right)^2 + \left(-\tfrac{1}{2}\right)^2 + 1^2} = \sqrt{\tfrac{1}{4} + \tfrac{1}{4} + 1} = \sqrt{\tfrac{3}{2}}.$$

So,

$$\mathbf{u}_2^* = \frac{1}{\sqrt{3/2}} \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right) = \left(\tfrac{1}{\sqrt{6}}, -\tfrac{1}{\sqrt{6}}, \tfrac{2}{\sqrt{6}}\right).$$

The orthonormal set (using $\mathbf{u}_1$ first) is

$$\left\{ \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right), \left(\tfrac{1}{\sqrt{6}}, -\tfrac{1}{\sqrt{6}}, \tfrac{2}{\sqrt{6}}\right) \right\}.$$

Alternative solution (ignoring the hint, starting with $\mathbf{u}_2$):

First normalize $\mathbf{u}_2$:

$$\|\mathbf{u}_2\| = \sqrt{1+0+1} = \sqrt{2}, \qquad \mathbf{v}_1' = \frac{\mathbf{u}_2}{\sqrt{2}} = \left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right).$$

Now orthogonalize $\mathbf{u}_1$ against $\mathbf{v}_1'$:

$$\langle \mathbf{u}_1, \mathbf{v}_1' \rangle = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) \cdot \left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right) = \tfrac{1}{2}.$$

$$\mathbf{w}' = \mathbf{u}_1 - \tfrac{1}{2}\mathbf{v}_1' = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) - \tfrac{1}{2}\left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right) = \left(\tfrac{1}{2\sqrt{2}}, \tfrac{1}{\sqrt{2}}, -\tfrac{1}{2\sqrt{2}}\right).$$

Simplify:

$$\mathbf{w}' = \tfrac{1}{2\sqrt{2}}(1,2,-1).$$

Its norm:

$$\|\mathbf{w}'\| = \sqrt{\tfrac{1}{8} + \tfrac{1}{2} + \tfrac{1}{8}} = \sqrt{\tfrac{3}{4}} = \tfrac{\sqrt{3}}{2}.$$

Normalize:

$$\mathbf{v}_2' = \frac{\mathbf{w}'}{\|\mathbf{w}'\|} = \frac{\frac{1}{2\sqrt{2}}(1,2,-1)}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{6}}(1,2,-1).$$

So the orthonormal set (using $\mathbf{u}_2$ first) is

$$\left\{ \left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right), \tfrac{1}{\sqrt{6}}(1,2,-1) \right\}.$$